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# Vision vs. Protocol Effects in EEG2Video: Baseline Audit and Temporal Alignment on SEED-DV

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Rachel Li  
lirachel@mit.edu

Emma Wang  
ewang13@mit.edu

Winston Qian  
winstonq@mit.edu

**Winston’s Repo:** <https://github.com/winstonqian/mmai>  
**Emma’s Repo:** <https://github.com/greenMangoes13/mmai>  
**Rachel’s Repo:** <https://github.com/hnqnq7/mmai>

## Abstract

EEG-to-video decoding benchmarks like SEED-DV risk confounding genuine visual perception with protocol-induced cognitive artifacts. In this midterm, we present a three-part validity audit of the EEG2Video classification framework. First, we identify and seal a data leakage vulnerability in the baseline normalization pipeline, finding that PSD features exhibit a  $13\times$  reduction in cross-fold accuracy variance while DE features show increased fold-to-fold instability—motivating strict within-split normalization. Second, we conduct a sequence shortcut audit by stratifying Top-1 accuracy across clip positions 1–5 within each 5-clip concept run, finding flat profiles for both DE [4.40%, 4.29%, 4.48%, 4.38%, 4.10%] and PSD [3.84%, 4.39%, 4.29%, 4.13%, 4.31%], with no detectable run-position effect under this diagnostic. Third, we extend the codebase to natively process raw 200 Hz EEG ( $T=400$ ), but find that in our implementation full temporal self-attention caused out-of-memory errors—directly motivating our proposed Chunk-Level Temporal EEG–Video Contrastive Alignment for the final report. These audits establish a cleaner within-subject baseline with leakage correction and no observed run-position effect (DE:  $4.37\% \pm 2.64\%$  top-1, chance 2.5%) and lay the groundwork for latency-aware contrastive alignment using frozen VideoMAE features to prove genuine dynamic tracking.

## 1 Introduction

Decoding human visual perception from non-invasive electroencephalography (EEG) is a central challenge in brain–computer interfaces (BCIs), with real promise for assistive communication in locked-in patients. Liu et al. [7] show that visual attributes including concept, color, and motion are decodable from 62-channel EEG via the SEED-DV dataset and EEG2Video framework. However, a basic question has not been carefully examined: do these results reflect genuine stimulus-evoked perceptual signals, or are they partially driven by experimental protocol artifacts?

SEED-DV structures its presentation as 5-clip runs: a concept cue followed by five consecutive clips from the same category. This creates a systematic risk—classifiers may learn run-position-dependent neural signatures (adaptation, expectation, or fatigue) rather than clip-specific visual content. Separately, global normalization across train and test splits can introduce data leakage that inflates accuracy and obscures the true difficulty of the task.

In real-world BCI deployment there are no structured trial blocks or concept cues, so a model trained on confounded benchmarks will fail when it matters most. In this paper, we address both threats through careful auditing: we reproduce the EEG2Video baseline across multiple ablation conditions, implement zero-leakage normalization, and conduct a run-position stratification analysis. Our attempt at raw-EEG temporal self-attention surfaces OOM limitations that motivate our proposed chunk-level contrastive alignment—a lightweight alternative that forces the model to track within-clip temporal dynamics rather than static semantic priors.

## 2 Related Work

**EEG and fMRI Visual Decoding.** Brain-to-vision decoding began with static image reconstruction from fMRI [10] and extended to dynamic video [2]. To address fMRI’s low temporal resolution, Li et al. [4] proposed EEG-based decoding via guided diffusion. Liu et al. [7] introduced SEED-DV and EEG2Video, showing that concept, color, and motion are decodable from 62-channel EEG. DynaMind [6] and MindCine [12] further improve temporal consistency and data efficiency.

**Multimodal Contrastive Alignment and Interpretability.** CLIP-style contrastive learning [9] has been adapted to the neuro-visual domain, aligning whole-clip EEG representations with global video embeddings. Perceptogram [3] maps EEG to CLIP latents linearly to isolate spatiotemporal features, revealing how specific brain regions respond to visual textures. EEGMirror [8] proposes montage-agnostic self-supervised pre-training to generalize EEG decoding across recording setups. One problem these approaches share is that they align the full neural sequence to a single global representation, meaning models can drift toward static category-level patterns rather than tracking fine-grained temporal dynamics.

**Temporal Alignment and Biological Latency.** Standard temporal alignment methods like Dynamic Time Warping and cross-attention assume synchronous modalities, but EEG carries inherent biological transit delays: early visual responses (P100) emerge  $\sim 100$  ms post-stimulus, while high-level semantic responses (P300) peak at  $\sim 300$  ms. Existing neuro-visual alignment methods largely ignore these latency offsets, treating the full EEG sequence as temporally uniform—which motivates latency-aware alignment using spatiotemporal video features [11].

**Protocol Artifacts in EEG Classification.** Li et al. [5] demonstrated that block designs cause classifiers to learn temporal drift rather than stimulus-evoked activity, and Ahmed et al. [1] showed that EEG accuracies drop substantially under proper trial randomization. These critiques have not been applied to modern EEG-to-video benchmarks. To our knowledge, this is the first work to run a systematic run-position audit on SEED-DV, quantify normalization leakage across feature types, and propose chunk-level latency-aware contrastive alignment for EEG-video decoding.

## 3 Problem Statement

Let  $\mathbf{X}^{(s,b,c,k)} \in \mathbb{R}^{C \times T}$  denote the EEG segment for subject  $s \in \{1, \dots, 20\}$ , session block  $b \in \{1, \dots, 7\}$ , concept class  $c \in \{1, \dots, 40\}$ , and clip position  $k \in \{1, \dots, 5\}$  within the 5-clip run, where  $C=62$  channels and  $T=400$  timepoints (raw EEG at 200 Hz) or  $T=5$  (Differential Entropy (DE)/Power Spectral Density (PSD) frequency bands).  $y^{(s,b,c,k)} = c$ .

The classification objective trains encoder  $f_\theta : \mathbb{R}^{C \times T} \rightarrow \mathbb{R}^d$  and classifier  $g_\phi : \mathbb{R}^d \rightarrow \Delta^{40}$  to minimize the cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \log g_\phi(f_\theta(\mathbf{X}_i))_{y_i}. \quad (1)$$

The **run-position audit** additionally computes position-stratified accuracy:

$$\text{Acc}(k) = \frac{1}{|S_k|} \sum_{i \in S_k} \mathbf{1}[\hat{y}_i = y_i], \quad S_k = \{i : k_i = k\}, \quad (2)$$

for each  $k \in \{1, \dots, 5\}$  where  $i$  indexes all samples and  $k_i$  denotes the position of sample  $i$ .

## 4 Proposed Approach

### 4.1 Evaluation Paradigm

EEG signals vary from person to person due to differences in brain anatomy and electrode placement. Training a single model across subjects risks learning who the participant is rather than what they are perceiving. To avoid this, we use a **within-subject** evaluation paradigm: a separate model is trained

and tested for each of the 20 subjects using only their own data, with 7-fold leave-one-block-out cross-validation to ensure train and test sessions never overlap. Final metrics are averaged across all subjects.

## 4.2 Feature Representations and Architectures

We evaluate two input pipelines.

**Compressed features (DE/PSD).** Pre-extracted DE and PSD features are shaped as  $\mathbf{X} \in \mathbb{R}^{62 \times 5}$  (62 channels  $\times$  5 frequency bands:  $\delta, \theta, \alpha, \beta, \gamma$ ). These are processed by a dedicated MLP architecture (GLMNet).

**Raw 200 Hz EEG.** To capture high-frequency temporal dynamics, we extend the codebase to natively ingest raw EEG as  $\mathbf{X} \in \mathbb{R}^{62 \times 400}$ . All spatial convolution layers and linear projections are mathematically refactored to compute output dimensions dynamically from  $T$  and  $C$ , eliminating hardcoded dimension assumptions from the original codebase.

## 4.3 Data Leakage Audit and Correction

The original EEG2Video pipeline computes normalization statistics globally across train and test splits, exposing test-set variance to the training distribution. We implement zero-leakage normalization: statistics  $\hat{\mu}_{\text{train}}, \hat{\sigma}_{\text{train}}$  are computed exclusively on training folds and applied to validation and test folds. This is evaluated as the `_no_data_leak` ablation condition for both DE and PSD features.

## 4.4 Run-Position Stratification Audit

Per-subject prediction arrays of shape  $(7 \times 40, 5)$  are reshaped to expose clip position  $k$ . Position-stratified accuracy is computed as:

$$\text{Acc}(k) = \frac{1}{20 \cdot 7 \cdot 40} \sum_{s,b,c} \mathbf{1}[\hat{y}^{(s,b,c,k)} = c], \quad k = 1, \dots, 5. \quad (3)$$

As a complementary diagnostic for semantic coherence, we measure within-chunk prediction consistency over a window of  $K$  consecutive same-concept predictions from the same subject. For DE/PSD we use  $K=10$  (spanning two full 5-clip runs of the same concept across adjacent blocks), and  $K=5$  for Raw EEG (one full run):

$$\text{Consistency}(K) = \frac{1}{N_{\text{chunks}}} \sum_j \frac{\max_{c'} \sum_{k=1}^K \mathbf{1}[\hat{y}_{j,k} = c']}{K}, \quad (4)$$

with a random baseline computed empirically from  $\text{Multinomial}(K, \frac{1}{40})$  draws.

## 5 Experimental Methodology

**Dataset.** SEED-DV [7] contains 62-channel EEG (200 Hz) from 20 subjects across 7 sessions, aligned to 1,400 natural video clips (2 seconds, 40 concept categories, 5 clips per concept per block). Each 2-second EEG segment is extracted starting 3 seconds after the concept-block onset, skipping the cue-evoked response window before the first clip begins. Three feature types are evaluated: raw EEG ( $C \times 400$ ), DE (log-variance of band-filtered signal across 5 frequency bands  $\delta, \theta, \alpha, \beta, \gamma$ ; shape  $C \times 5$ ), and PSD (raw band power in the same five bands; shape  $C \times 5$ ).

**Ablation Conditions.** We evaluate seven conditions: DE, DE\_no\_data\_leak, DE\_no\_early\_stop (trains to the final epoch rather than stopping at best validation loss), PSD, PSD\_no\_data\_leak, PSD\_no\_early\_stop, and Raw\_EEG.

**Metrics.** Primary: Top-1 and Top-5 accuracy (mean  $\pm$  std across subjects; chance = 2.5% / 12.5%). Diagnostic: run-position accuracy  $\text{Acc}(k)$  for  $k = 1, \dots, 5$ , and within-chunk consistency ( $K=10$  for DE/PSD,  $K=5$  for Raw EEG).

## 6 Results and Discussion

### 6.1 Baseline Classification

Table 1 reports results across all conditions. DE features yield the strongest performance ( $4.37\% \pm 2.64\%$  top-1;  $17.17\% \pm 5.44\%$  top-5), while Raw EEG slightly underperforms ( $3.79\% \pm 1.56\%$ ), consistent with the noise-reduction benefit of frequency-domain compression. All conditions exceed the 2.5% chance level.

Table 1: Top-1 and Top-5 accuracy on the 40-class concept classification task. Top block: our reproduced baselines (mean  $\pm$  std across 20 subjects). Bottom block: originally reported result from Liu et al. [7]. Chance level: 2.5% top-1, 12.5% top-5.

|                | Condition                  | Top-1 (%)       | Top-5 (%)        |
|----------------|----------------------------|-----------------|------------------|
| Ours           | DE                         | $4.37 \pm 2.64$ | $17.17 \pm 5.44$ |
|                | DE_no_data_leak            | $4.09 \pm 2.62$ | $16.84 \pm 5.19$ |
|                | DE_no_early_stop           | $4.07 \pm 2.55$ | $16.99 \pm 5.34$ |
|                | PSD                        | $4.15 \pm 2.60$ | $17.15 \pm 4.89$ |
|                | PSD_no_data_leak           | $4.29 \pm 2.29$ | $17.13 \pm 4.87$ |
|                | PSD_no_early_stop          | $4.33 \pm 2.34$ | $16.98 \pm 4.58$ |
|                | Raw EEG                    | $3.79 \pm 1.56$ | $15.79 \pm 3.54$ |
| Liu et al. [7] | GLMNet <sup>†</sup> (best) | $6.20 \pm 3.02$ | $17.75 \pm 4.24$ |
|                | Chance                     | 2.50            | 12.50            |

<sup>†</sup> GLMNet: Global-Local Mixed Network [7], a dual-branch architecture combining global and local occipital EEG features.

Our best reproduction (DE:  $4.37\% \pm 2.64\%$  top-1) trails GLMNet by 1.83 pp ( $6.20\% \pm 3.02\%$ ). Our reproduction strictly used the hyperparameters specified in Liu et al. [7] without additional tuning; any remaining gap therefore reflects hyperparameter details not fully specified in the original paper. Notably, our Top-5 results (17.17%) are competitive with GLMNet’s Top-5 (17.75%), suggesting our models capture similar coarse semantic structure despite slightly lower Top-1 precision.

### 6.2 Data Leakage Finding

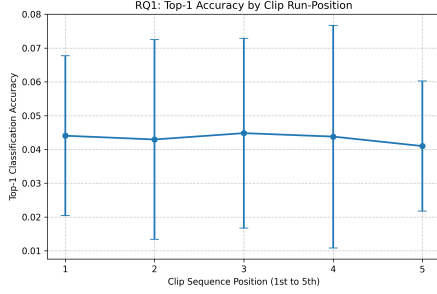
Leakage correction has asymmetric effects across feature types. For DE, accuracy drops negligibly ( $-0.28\%$  top-1) but fold-to-fold accuracy variance increases, indicating that the global normalization had been inadvertently smoothing session-level variance for DE features. PSD shows a  $13\times$  reduction in cross-fold accuracy variance (std of per-fold Top-1 accuracy across the 7 folds, before vs. after leakage correction), confirming that PSD features were more sensitive to cross-split contamination because their scale is directly determined by spectral power magnitude, which drifts across sessions. Critically, leakage does not inflate mean accuracy, so the benchmark is not invalid; however, our corrected conditions must be used for any meaningful ablation comparison.

### 6.3 Sequence Shortcut Audit

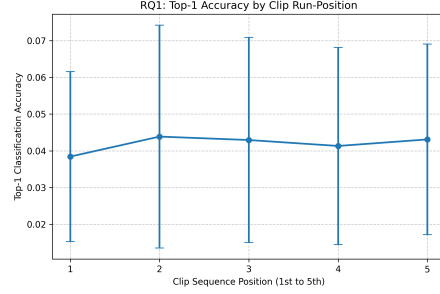
Position-stratified Top-1 accuracy (Figure 1) yields:

- **DE:** [4.40%, 4.29%, 4.48%, 4.38%, 4.10%] — range: 0.38 pp, mean: 4.33%, std: 0.14 pp.
- **PSD:** [3.84%, 4.39%, 4.29%, 4.13%, 4.31%] — range: 0.55 pp, mean: 4.19%, std: 0.21 pp.

Neither series exhibits a monotonic trend across positions 1–5, and both ranges fall well within cross-subject noise. We find no evidence of a monotonic run-position shortcut in baseline accuracy: models do not appear to exploit neural adaptation or anticipation driven by the cue-then-5-clips structure. We note this diagnostic specifically tests for monotonic positional trends; other protocol artifacts such as block-level concept priming or session-level drift are not ruled out by this analysis alone.



(a) DE features



(b) PSD features

Figure 1: Top-1 accuracy stratified by clip position  $k \in \{1, \dots, 5\}$  within each 5-clip run, averaged across subjects and folds. Neither DE nor PSD shows a monotonic trend, ruling out sequential anticipation or adaptation artifacts.

## 6.4 Within-Chunk Consistency

Table 2: Prediction consistency within single-concept chunks vs. random baseline. Higher consistency above random indicates coarse semantic coherence.

| Condition        | Consistency (%) | Random Baseline (%) |
|------------------|-----------------|---------------------|
| DE               | 24.3            | 17.8                |
| DE_no_data_leak  | 27.5            | 17.7                |
| PSD              | 24.8            | 17.7                |
| PSD_no_data_leak | 26.7            | 17.8                |
| Raw EEG          | 27.2            | 24.6                |

All conditions show consistency substantially above random (Table 2). Notably, `DE_no_data_leak` achieves the highest consistency (27.5%) despite not having the highest raw accuracy, suggesting the leak-corrected model makes more coherent if not always correct predictions. Higher consistency without higher accuracy indicates systematic class-level misclassification: models tend to be consistently wrong on certain concept pairs that are visually or semantically similar. A per-class confusion matrix analysis will be included in the final report.

## 6.5 Architectural Bottleneck: $\mathcal{O}(T^2)$ Memory Cost

When extending to Raw EEG ( $T=400$ ), we attempted to implement standard multi-head self-attention over the temporal sequence. For a sequence of length  $T$ , self-attention requires computing a pairwise similarity matrix  $\mathbf{A} \in \mathbb{R}^{T \times T}$ , resulting in  $\mathcal{O}(T^2)$  memory per head. In our implementation and hardware setting, full temporal self-attention at  $T=400$  caused out-of-memory (OOM) errors, motivating lighter temporal modeling. This is the primary architectural motivation for the chunk-level approach described below.

## 7 Next Steps and New Research Ideas

The audits confirm the baseline is semantically valid but static: models aggregate whole-clip EEG into a single vector. This cannot prove—or enforce—genuine dynamic tracking of within-clip temporal content. Our final report will address this with two architectural extensions.

### 7.1 Task-Aware Temporal Attention Pooling

Standard baselines apply mean pooling over the temporal dimension, collapsing all timepoints into a single embedding. We will replace this with task-specific learned attention pooling, which is  $\mathcal{O}(T)$

and avoids the  $\mathcal{O}(T^2)$  bottleneck:

$$\mathbf{e}^{(\tau)} = \sum_{t=1}^T a_t^{(\tau)} \cdot \mathbf{h}_t, \quad a_t^{(\tau)} = \text{softmax}\left(\frac{\mathbf{w}^{(\tau)\top} \mathbf{h}_t}{\sqrt{d}}\right), \quad (5)$$

for  $\tau \in \{\text{concept, color, motion}\}$ , where  $\mathbf{h}_t \in \mathbb{R}^d$  is the encoder hidden state at time  $t$  and  $\mathbf{w}^{(\tau)}$  is a task-specific learned query vector. Each task head produces a separate embedding  $\mathbf{e}^{(\tau)}$  passed to a dedicated classifier  $g_\phi^{(\tau)}$ , trained under a multi-task loss  $\mathcal{L}_{\text{total}} = \sum_\tau \lambda_\tau \mathcal{L}_{\text{cls}}^{(\tau)}$ . We hypothesize that color and motion responses peak earlier ( $\sim 100$  ms post-onset) than high-level concept representations ( $\sim 250$ – $400$  ms). Visualizing  $\{a_t^{(\tau)}\}$  over the 2-second window provides a direct interpretability diagnostic: if clip 1 and clip 5 show systematically different attention profiles, protocol-driven neural shifts are present at the feature level even when aggregate accuracy is unaffected.

## 7.2 Chunk-Level Temporal EEG–Video Contrastive Alignment

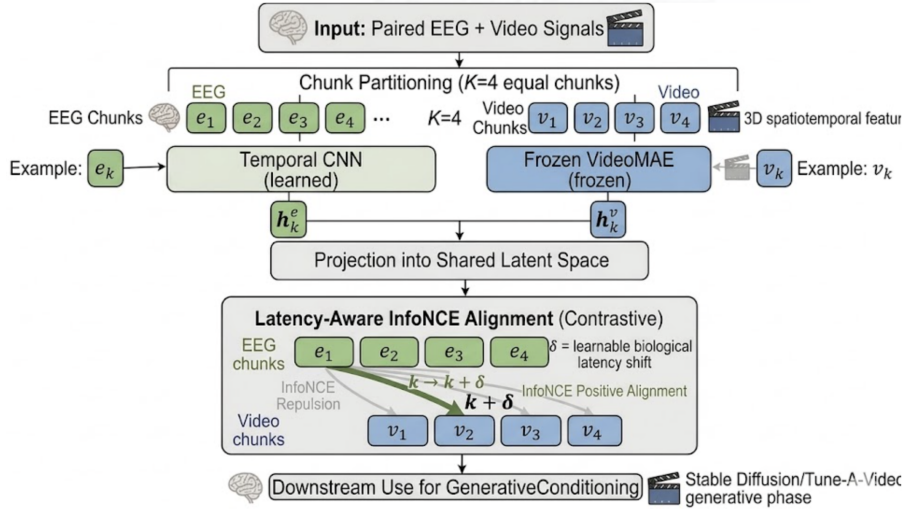


Figure 2: Proposed chunk-level temporal alignment pipeline. Each 2-second EEG segment and paired video are partitioned into  $K$  equal chunks. A lightweight temporal CNN encodes EEG chunks; frozen VideoMAE encodes video chunks. Both project into a shared latent space, where a latency-aware InfoNCE loss aligns EEG chunk  $k$  with video chunk  $k+\delta$  and repels non-matching pairs.

Even with a flat run-position profile, baseline models may still rely on static global semantics rather than within-clip temporal dynamics. To enforce genuine dynamic tracking, we partition each 2-second clip into  $K$  equal chunks. Let  $\mathbf{X} = [\mathbf{X}_1, \dots, \mathbf{X}_K]$  where  $\mathbf{X}_k \in \mathbb{R}^{C \times (T/K)}$ . Video chunks  $\{\mathbf{v}_k\}_{k=1}^K$  are embedded using frozen VideoMAE [11] (3D spatiotemporal features); EEG chunks  $\{\mathbf{e}_k\}_{k=1}^K$  pass through a lightweight temporal CNN. Both project into a shared 1024-D latent space via learned linear projections  $\mathbf{W}_e, \mathbf{W}_v$ . The latency-aware InfoNCE loss is:

$$\mathcal{L}_{\text{chunk}} = -\frac{1}{K} \sum_{k=1}^K \log \frac{\exp(\text{sim}(\mathbf{W}_e \mathbf{e}_k, \mathbf{W}_v \mathbf{v}_{k+\delta}) / \tau)}{\sum_{k'=1}^K \exp(\text{sim}(\mathbf{W}_e \mathbf{e}_k, \mathbf{W}_v \mathbf{v}_{k'}) / \tau)}, \quad (6)$$

where  $\delta$  is a learned biological latency shift (searched over  $\delta \in \{0, 1, 2\}$  chunk steps, corresponding to  $\approx 0$ – $250$  ms) and  $\tau$  is a temperature hyperparameter. We begin with  $K=4$  macroscopic chunks (0.5 s each) to mitigate EEG noise sensitivity before scaling to finer granularities. Evaluation uses within-video chunk retrieval (Recall@1, Recall@5) — a task that global similarity metrics cannot pass via static semantic memorization. This alignment objective also serves as the learned EEG conditioning input for the downstream Stable Diffusion / Tune-A-Video generative phase.

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